

What Makes DirectX Tick?

DirectDraw: Allows direct access to areas in the display adapter's video memory and hardware buffers for tasks such as page flipping (used while creating animations to display one screen while the next is being created). Allows programs to make full use of the colour decompression capabilities of various types of hardware and not be dependent on a specific type of hardware.

Direct3D: Allows access to the advanced features that handle 3D functions such as triangle creation, scaling and transforming 3D images.

This component is scalable, allowing part or all of the 3D rendering pipeline access to more advanced features such as z-buffering and anti-aliasing.

Direct Input: Implements support for analog and digital input devices, offering increased responsiveness and sensitivity.

DirectSound: Offers direct access to the audio sub-system. Enables features such as low-latency mixing (where multiple audio sources are used with the soundcard) and hardware acceleration for high sampling rates when many audio streams are used, like in the new breed of games that have surround-sound.

DirectPlay: The networking component, which provides access to devices over a network—independent of the hardware, protocols and networking services that are used.

DirectMusic: Part of the direct sound sub-system that streamlines the reproduction of music such as MIDI playback in applications and games that are written to use it.

GEFORCE3

Graphics hardware has really matured over the past couple of years. The performance capabilities of today's 3D accelerators let game developers truly display their artistic skills. The emphasis undoubtedly is on realism.



You want to be able to see the ripples on that stream when you're walking through the fantasyland of your favourite role-playing game, and manufacturers such as nVidia have been consistently churning out the hottest gaming hardware to bring these effects to life.

nVidia products usually score both in terms of sheer performance and their use of innovative technologies. So does the GeForce3 carry on the tradition, or is it just another graphics card churned out from the assembly lines with the mandatory increase in specifications? Let's take a look at some of its features and then decide.

The nfiniteFX engine

One of the major enhancements in GeForce3's technology is the ability to allow game developers to specifically program certain parts of this graphics processor, allowing it to deliver custom special effects. These technologies are collectively referred to by nVidia as the nfiniteFX engine. Two important components of this new architecture are the Programmable Vertex Shader and the Programmable Pixel Shader.

Programmable Vertex Shader: In any 3D model, the vertex is the most basic element. This is the point at which the polygons in a 3D model meet, so any changes in the position and other characteristics of a vertex can change the appearance of the 3D model. Until now, vertices could be assigned to perform a very limited number of operations in the 3D application. However, with the GeForce3 there are a number of custom



The Programmable Vertex Shader will allow you to see wrinkles on faces in real-time



The Programmable Pixel Shader will let you see combined effects like reflective bump mapping

special effects, such as morphing, animation, deformation and environmental effects, which can be applied to vertices. This gives programmers a high amount of flexibility in what can be done to the 3D objects in the game.

Effects such as fog, lens effects and water refraction are now possible. Also, effects such as wrinkles and stretch marks can be created on the faces of characters in games, objects can morph into other elements, and mechanical processes such as gear trains can be simulated accurately. This will mean more natural animations for next-generation game characters.

Programmable Pixel Shader: Like the Programmable Vertex Shader, this feature allows application developers to take one step further towards creating life-like 3D scenes and effects. This technology has been around since the GeForce2 was launched. It was then known as the nVidia Shading Rasteriser or NSR. Using it, effects such as reflections, bumps, fog and textures could be applied at the pixel-level in a scene. The GeForce3 processor takes this to a much higher level by allowing up to eight texture operations in a single pass. This is coupled with the ability to handle four textures at a time. The GeForce3 can combine these operations to produce breathtaking special effects such as reflective bump mapping and realistic textures with environmental effects such as fog and mist. Since these effects are processed in parallel, it does not affect the frame rates in games, but it certainly makes games look a whole lot better!

Lightspeed Memory Architecture

nVidia graphics cards have always been notorious for problems with their memory bandwidth. As 3D processors keep getting faster, even though exotic memory like DDR is used, performance is hit at higher resolutions because of the graphics card's memory architecture.